Monica Seglar-Arroyo

Lead Research Scientist & Systems Architect

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PROFESSIONAL SUMMARY

Strategic Scientific and Technical Leader and Principal Software Architect with 10+ years of extensive experience directing high-impact data analysis and engineering, as well as infrastructure projects within premier global consortia. Proven capability in managing cross-functional international teams (+50-person teams across 6 time zones), designing novel optimization systems and optimizing existent ones, reducing operational delays up to by 50%, translating complex statistical theory into resilient, production-grade enterprise software architectures and securing competitive R&D funding. Currently seeking quantitative analyst or senior data science role where mathematical rigor drives business impact.

CORE COMPETENCIES

TECHNICAL ARCHITECTURE

- Big Data Architecture & Analysis
- Real-Time Time Series Analysis
- Automated CI/CD Pipelines
- Advanced Statistical Modeling
- Heuristic & Deterministic Optimization
- Genetic Algorithms & Machine Learning
- R&D AI Tooling & Prompt Engineering

LEADERSHIP & GOVERNANCE

- International Project Management
- Multidisciplinary Team Leadership (+50)
- Operational & Predictive Risk Management
- Strategic Alliance & Consortium Building
- Global Stakeholder Negotiation
- Ph.D. Researcher Supervision & Mentorship
- Technical Governance & Global Dissemination

PROFESSIONAL EXPERIENCE

Institut de Fisica d'Altes Energies (IFAE) | Barcelona, Spain
Senior Postdoctoral Researcher/ Lead Systems Architect | Nov 2022- Jan 2026

- Deployed a production-grade optimization engine (tilepy) as the official orchestrator and scheduling motor for the global Big Data infrastructure of global consotium. Managed multi-million euro resources under incomplete information, achieving a 60%+ reduction in operational delays.
- Design of tileX as a new approach for solving NP-hard problems, applying genetic algorithms oriented toward resource allocation. Development of predictive analytical frameworks on distributed architectures in high-performance computing environments, identifying strategic opportunities and uncertainties within top-tier global consortia.
- Managed high-stakes operations under critical minute decision windows, establishing coordination processes adopted as organizational standards, while improving critical measurement infrastructure precision and reducing latency for data utilized by 100+ professionals globally, contributing to a broader scientific body of work recognized in 27K+ research citations.
- Co-supervised/mentored 3/6 PhD researchers. Authored and audited high-impact proposals to secure international funding for resource allocation, AI initiatives. Articulated inter-institutional networks, and defended the strategic positioning of results at over 40 international conferences.

Laboratoire d'Annecy de Physique des Particules (LAPP)| Annecy, France
Postdoctoral Researcher/ Core Software Engineer | Nov 2019-Oct 2022

- Directed the architectural redesign, core development, and packaging of the open-source framework tilepy (GPL3), engineered to solve dynamic resource scheduling for distributed hardware fleets (NP-hard, Best-Next approach).
- Developed and validated core analytical streaming pipelines for high-dimensional spatiotemporal signal reconstruction under extreme noise conditions. Co-led the architecture of low-latency feedback loops for hardware actuators, integrating automated analytical models for real-time noise subtraction and bias correction.
- Co-managed the analytical ingestion and integration of distributed, highly heterogeneous Big Data pipelines across multiple international consortia. Designed automated anomaly and regime-change detection systems for high-volatility, continuous big data streams using advanced statistical techniques and Bayesian Block analysis.
- Leadership in defining technical documentation standards and software architecture manuals for auditability. Implementation of CI/CD pipelines.

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EDUCATION

PhD in High Energy Physics | Université Paris-Saclay, France | 2016 - 2019

- Thesis: Studying the Origin of Cosmic-Rays : Multi-Messenger studies with Very-High-Energy Gamma-Ray Instruments
- Visiting Scholar: Penn State University, USA (Aug 2017 - Jul 2018)

MSc in Fundamental Physics | Université Grenoble-Alpes, France | 2015-2016

- Thesis: Improving the Precision of H.E.S.S. Data Analysis via Advanced Atmospheric Profiling
- Distinction: Full Excellence Scholarship (LabEx ENIGMASS)

BSc in Physics | Universidad de Valencia, Spain | 2010-2015

- International Experience: Erasmus Student at University of Bonn, Germany (2012-2013)

KEY SKILLS & QUALIFICATIONS

Technical Stack & Tooling

- Language & Ecosystems: Python, C/C++, LaTeX, Bash
- DevOps & Infrastructure: GitLab CI/CD, GitHub, Condor HPC, PyPI
- Frameworks & Libraries: Scikit-Learn, TensorFlow, NumPy, SciPy, Pandas. tilepy, tileX as creator

Certifications & Professional Development

- Generative AI with LLMs | DeepLearning.AI | in progress
- Investment Management Specialization | University of Geneva & UBS (coursera) | in progress
- From Science to Business | ESADE Business School | 2024

Languages

- Spanish - Native
- Catalan - Native
- English - C1
- French - C1
- Italian - A2
- German - A2

SIDE PROJECTS

Co-Founder & Strategy Lead | VERT Adventures, Barcelona | Sep 2025 - Present

- Founded and scaled outdoor adventure company (€40K+ annual revenue, 100+ customers in first year) while maintaining full-time research career.
- Managed end-to-end operations; translate analytical skills into revenue-generating outcomes

ML consultant/expert | UJI and CAPSBE | 2026-Present

- Lead the study and algorithmic optimization of nursery resource allocation based on multi-morbidity indices using advanced Machine Learning models (TFR project).